CS 569 Machine Learning

Homework 5: K-means Clustering

1 Objective

By completing this assignment, you will:

- Understand and implement the K-means clustering algorithm.
- Learn how to evaluate cluster quality using SSE.
- Apply the Elbow Method to determine an optimal K.
- Compare your custom implementation to Python's built-in version.

2 Dataset

You will be provided with a .txt file containing the dataset. Each row represents a data point, and each column represents a feature.

3 Tasks

3.1 Load the dataset into Python

Print the shape and the first 5 rows to confirm correct loading.

3.2 Implement K-Means From Scratch

You are NOT allowed to use any clustering libraries for this part.

Your implementation must include:

- Initialize Centroids:
  - Randomly select K points from the dataset as initial centroids.
- Assign Points to Clusters:
  - For each data point, compute the Euclidean distance to each centroid.
  - Assign the point to the nearest centroid.
- Update Centroids:
  - For each cluster, calculate the new centroid as the mean of all points assigned to that cluster.
- Check for Convergence:
  - Stop iterating when centroids do not change significantly (e.g., centroid movement $< 1e-4$)
  - OR after a maximum number of iterations (e.g., 300).
- Compute SSE:
  - For each cluster, calculate the sum of squared distances of points to their centroid.
  - Total SSE is the sum across all clusters.

3.3 Use the Elbow Method

- Run your implementation for $K = 1$ to $K = 10$.
- Plot SSE vs. K.

- Identify the elbow point visually, i.e., the point where SSE starts decreasing at a slower rate. As shown in Fig 1

- Suggest an optimal K based on your plot.

3.4 Run Scikit-Learn's Built-in KMeans

- Using klearn.cluster.KMeans, run the algorithm on the same dataset with the same K.

- Compare:

  - Final centroids

  - SSE

  - Cluster assignments

This would need to reflect the degree to which the indivual objects cluster together as the labels assigned are arbitrary. The approach would need to be built around the degree to which the same objects cocluster

- They should be very similar, though not always identical, due to random initialization.

3.5 Visualize the Clusters

- Color each point by its cluster assignment. As shown in Fig 2

- Mark centroids with a distinct shape (e.g., 'X').

4 Items to submit

A zip file that contains your Python source code and a PDF report.

5 Grading Rubric: K-Means Clustering Assignment

Criteria	**Points**
Correct K-Means implementation	60
Proper use of Elbow Method	10
Comparison with scikit-learn KMeans	10
Visualizations	15
Brief report quality	5
Total	**100**

Table 1: Grading rubric for the K-Means clustering assignment.

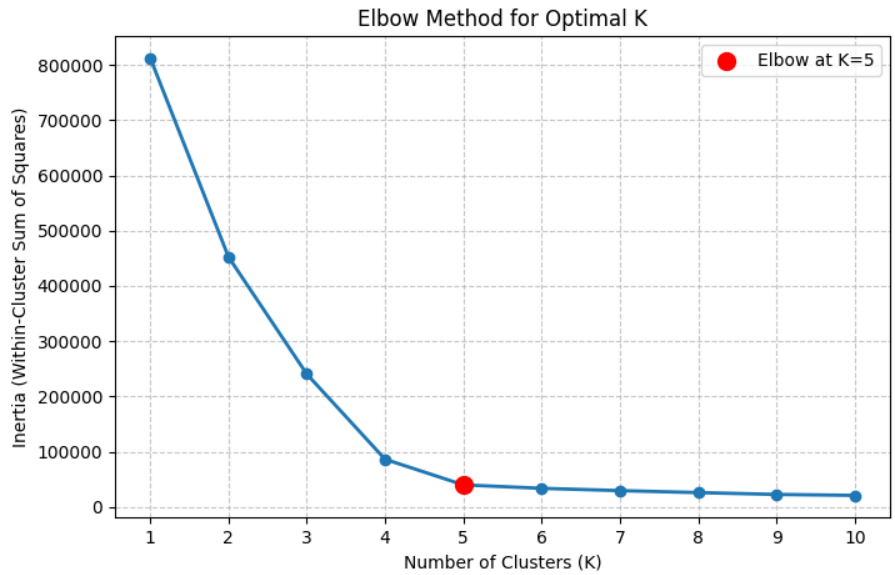

Figure 1: Elbow Method

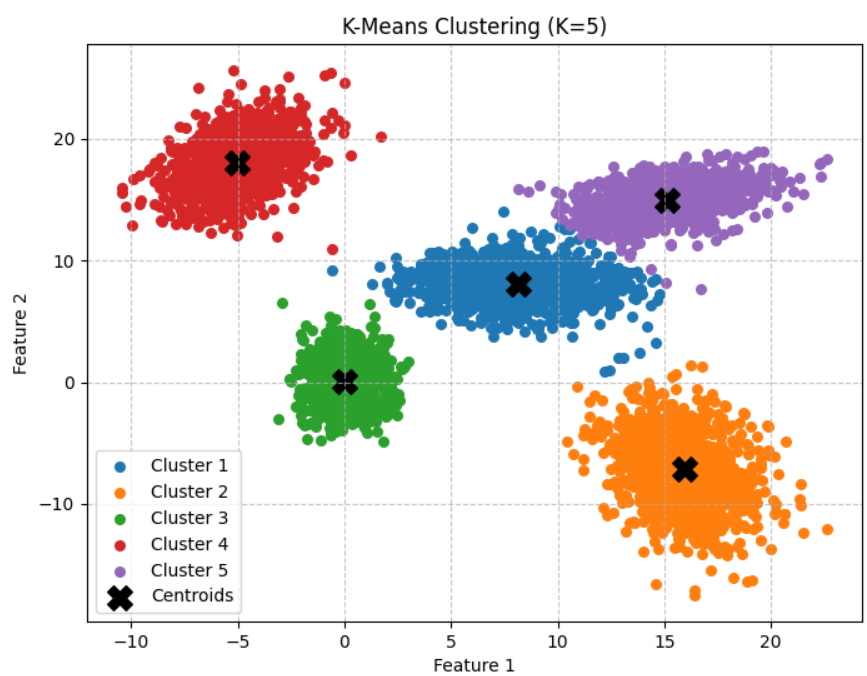

Figure 2: Clustering Result